
photon-photon physics at the LHC

for pedestrians

Laurent Schoeffel

Unless stated explicitly, we write all formulae in natural units

$$\epsilon_0 = \mu_0 = c = \hbar = 1$$

Only in a few relations, c is written explicitly

Chapter I

Elastic (protons) scattering

$$p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + X$$

Motivation

In general, at the LHC, we consider reactions originating from the interactions of quarks and gluons from both protons. This leads to cross sections of the form:

$$\sigma(p + p \rightarrow X) = \int \int q(x_1; \mu^2) \bar{q}(x_2; \mu^2) \sigma_{q\bar{q} \rightarrow X}(x_1, x_2; \mu^2) dx_1 dx_2$$

where μ^2 is a non-perturbative scale. There are some complications to this expression:

- Perturbation theory
- Factorization proved only for specific observables in specific kinematics
- Interactions between spectator partons are non perturbative
- These effects do not matter only in sufficiently inclusive observables
- Parton showers...

A lot of this complexity is hidden in MCs...

In the same spirit as: $\sigma(p + p \rightarrow X) = \int \int q(x_1; \mu^2) \bar{q}(x_2; \mu^2) \sigma_{q\bar{q} \rightarrow X}(x_1, x_2; \mu^2) dx_1 dx_2$, the idea (in the following) is to search for reactions initiated by photons emitted by both protons, such that we can write:

$$\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + X) = \int \int f(\omega_1) f(\omega_2) \sigma_{\gamma\gamma \rightarrow X}(\omega_1, \omega_2) \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}$$

where $f(\omega)$ are the equivalent to the parton densities, here for photons emitted by both protons (ω is the frequency or energy of the photon -we take $\hbar = 1$ unless stated explicitly-). Next, we show:

- how to determine the function(s) $f(\cdot)$ and why there is factorization
- why there is no scale appearing
- why this last formula needs to be improved...

Then, pp (or $PbPb$) collisions can be transformed into photon-photon collisions, with much less impact from non-perturbative effects. However, this makes sense only if these reactions are sizable at the LHC.

Charge q moving at constant $v = v_z$

- What is the EM field $(\vec{E}, \vec{B})$ of a charge q moving at constant velocity $\vec{v} = v\vec{e}_z$?
- Obviously, in the reference frame R' linked to the charge (where q is at rest), $\vec{E}'(\vec{r}')$ is the Coulomb field (in $\vec{r}'$ coordinates) and $\vec{B}'(\vec{r}')$ is zero.
- What is needed is to make the Lorentz transformation of the EM field from R' to the reference frame R where q is moving at velocity v (along (Oz)): $(\vec{E}'(\vec{r}'), \vec{B}'(\vec{r}')) \rightarrow (\vec{E}(\vec{r}), \vec{B}(\vec{r}))$.

We have:

$$E_z(x, y, z, t) = \frac{q}{4\pi} \gamma \frac{z - vt}{[\gamma^2(z - vt)^2 + x^2 + y^2]^{\frac{3}{2}}},$$

where $[\gamma^2(z - vt)^2 + x^2 + y^2]^{\frac{3}{2}}$ is equal to r'^2 the radial coordinate expressed in R' . Also,

$$E_x(x, y, z, t) = \frac{q}{4\pi} \gamma \frac{x}{[\gamma^2(z - vt)^2 + x^2 + y^2]^{\frac{3}{2}}},$$

$$E_y(x, y, z, t) = \frac{q}{4\pi} \gamma \frac{y}{[\gamma^2(z - vt)^2 + x^2 + y^2]^{\frac{3}{2}}}.$$

Charge q moving at constant $v_z \simeq c$

And, for the magnetic field in reference frame R : $\vec{B} = \vec{v} \times \vec{E}$.

Clearly, $\vec{E}(\vec{r})$ is a radial field in the reference frame R (as trivially in R').

Proposition 1. *The EM field $(\vec{E}, \vec{B})$ of a charge q moving at constant $v_z \simeq c$ is like the field of an EM wave.*

Proof. Perpendicular to the direction of motion: $z - vt = 0$, we have:

$$E = \sqrt{E_x^2 + E_y^2} = \frac{q}{4\pi} \gamma \frac{1}{[x^2 + y^2]},$$

which is like a Coulomb field but larger due to the Lorentz factor $\gamma > 1$. In front of the charge (or before) at $x = y = 0$, we have: $E = E_z = \frac{q}{4\pi} \frac{(1-v^2/c^2)}{(z-vt)^2}$, which is also like the Coulomb field but reduced by the factor $(1 - v^2/c^2)$.

When $v \simeq c$, the $\vec{E}$ is then (approximately) transverse, like for an EM wave, and $\vec{B} = \vec{v} \times \vec{E}$ is exactly transverse ($B_z = 0$). $\square$

Charge q moving at constant $v_z \simeq c$

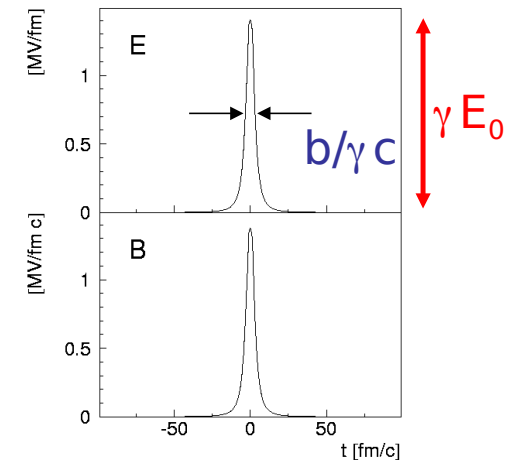
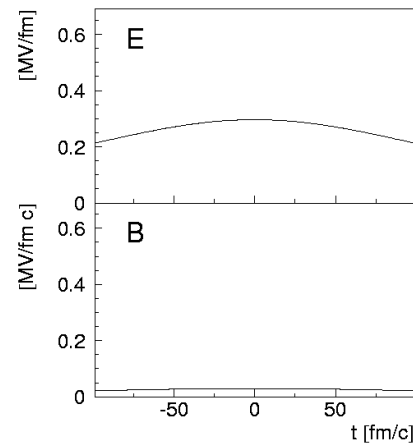
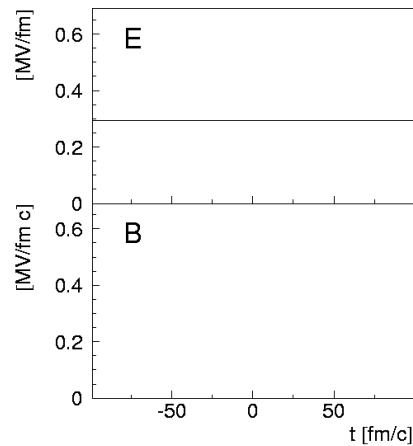
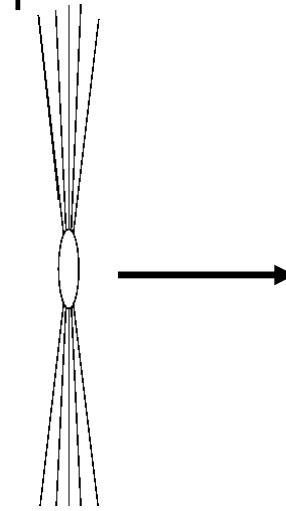
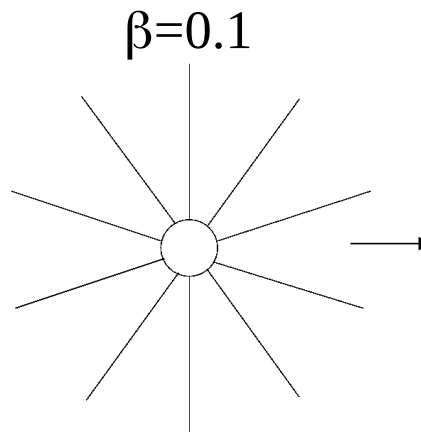
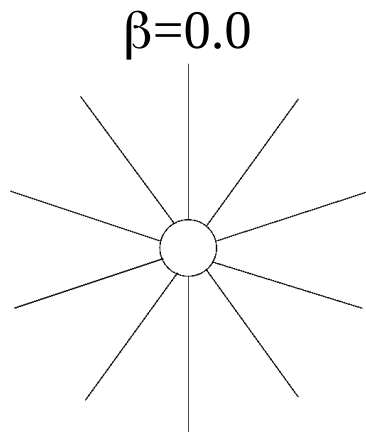
Proposition 2. *The EM field $(\vec{E}, \vec{B})$ produced by the charge q moving at $v_z \simeq c$ acts like a laser pulse.*

Proof. Let us express this field in the plane of equation $z = 0$ at a transverse position $\vec{b} = (x, y)$ (transverse to the direction of motion):

$\vec{E}(x, y, z = 0, t) = \vec{E}_T = \frac{q}{4\pi} \gamma \frac{\vec{b}}{[\gamma^2(vt)^2 + b^2]^{\frac{3}{2}}}$ and $\vec{B} = \vec{B}_T = \vec{v} \times \vec{E}_T$. Moreover, E_z is almost zero (see previous slide), while B_z is exactly zero.

When the time t is increasing, $vt \simeq ct$ is getting larger and the charge q is getting away from the plane $z = 0$, then γct becomes much larger than b and E_T is reduced by a factor $1/\gamma^2 \ll 1$. Therefore, this field $(\vec{E}, \vec{B})$ is non negligible only during an interval of time such that $\gamma c \delta t \simeq b$, which gives: $\delta t \simeq \frac{b}{\gamma c}$. The distance from the plane $z = 0$ is then $c\delta t \simeq \frac{b}{\gamma}$. This means: (1) b (also called the impact parameter) must be large; (2) in any transverse plane, the EM field produced by the charge q moving at $v_z \simeq c$ acts like a laser pulse of order $\delta t \simeq \frac{b}{\gamma c}$. $\square$

$R(\text{Au,Pb}) \approx 7 \text{ fm}$ An observer at a distance of 20 fm $\beta=0.98$



Large impact parameter

From $\vec{E}(x, y, z = 0, t) = \vec{E}_T = \frac{q}{4\pi} \gamma \frac{\vec{b}}{[\gamma^2(vt)^2 + b^2]^{\frac{3}{2}}}$, we have shown that b (the impact parameter) needs to be large for a non-negligible EM field.

In the case of the proton:

- b needs to be larger than R_p , the size of the proton. This is a first condition of coherence.
- The property of existence of the field (during a short interval of time): $\delta t < \frac{R_p}{\gamma c}$ will be translated (later) into another condition of coherence on the maximum energy of the photons.

In particular, for a pp collision where each acts as a (pulsed) source of real photons, the interaction will not happen inside the protons. But at least the conditions $b_1 > R_p$ and $b_2 > R_p$ must be satisfied. This is also known as the condition for peripheral collisions.

Interestingly: In general, all reactions that are labeled as diffractive interactions are known to be peripheral. But this is another story...

Notes:

- As the EM field is like the field of EM wave, we understand (intuitively) that photons emitted by the charge q (moving at $v_z \simeq c$) are real (or quasi-real). This will be shown.
- We have introduced (mathematically) some properties of coherence. These are obviously the conditions for the protons to stay intact during the pp interaction of interest here:
 $p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + X$. We will also come back later on these ideas. *They need to be distorted at some point.*

Remark

$E_{(x,y)}(x, y, z, t) = \frac{q}{4\pi} \gamma \frac{(x,y)}{[\gamma^2(z-vt)^2 + x^2 + y^2]^{\frac{3}{2}}}$, is the Electric field for the components (x, y) , expressed at point (x, y, z) in space. We can remark that it depends on $(x, y, z - vt)$, which are the coordinates measured for the **actual** position of the particle (located at $z = vt$).

We could have expected that it would depend on $z - vt'$, with $t' = t - r'/c$, which is the **retarded** time that matters for the field measured at time t (and r' the retarded position).

The solution is that the particle is moving on a straight line, therefore there is no problem here. And this formula is perfectly correct.

More generally, if we make the hypothesis that the potentials (of EM) depend only on the position and the velocity at the retarded time, whatever the movement of the particle,

the potentials of a charged particle moving at constant velocity are sufficient to derive everything (for any motion).

From the previous slides, we can already write an essential extension of the original formula:

$$\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + X) = \int \int f(\omega_1) f(\omega_2) \sigma_{\gamma\gamma \rightarrow X}(\omega_1, \omega_2) \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}$$

Indeed, we can not include the peripheral requirement in this simple expression, as long as the impact parameters do not appear explicitly. Therefore, a more general formula follows from the replacement: $f(\omega_1) f(\omega_2) \rightarrow \int \int n(\vec{b}_1, \omega_1) n(\vec{b}_2, \omega_2) d^2\vec{b}_1 d^2\vec{b}_2$.

Then, any constraint on the impact parameters can be imposed.

Important note: we have not proved yet that this formula makes sense, in the sense that photon distributions can be factorized with the microscopic $\gamma\gamma$ cross section -it will be done later-.

Next: from the EM field (derived in the previous slides), we will compute the function: $n(\vec{b}, \omega)$.

Equivalent photon number

For EM fields, it is always possible to compute the (equivalent) photon spectrum.

Indeed, the flux of the field is provided by the Poynting vector: $\vec{P}(\vec{r}, t) = \vec{E}(\vec{r}, t) \times \vec{B}(\vec{r}, t) \simeq E_T^2(\vec{r}, t) \vec{e}_z = E_T^2(\vec{b}, z, t) \vec{e}_z$.

Next, let us pose $n(\omega)$ (with ω the photon frequency -energy-), the (equivalent) number of photons is (generically) defined as:

$$\int dt \int d\vec{S} \vec{P}(\vec{b}, z, t) = \int_0^\infty d\omega \omega n(\omega)$$

The energy flux of the EM fields through a transverse plane is identical to the energy flux of the equivalent photons.

For simplicity in the following, we extract the 4π from the expressions of the EM fields.

We need to be careful that the integral on ω is defined only from 0 to ∞ .

All in, we obtain (per unit frequency): $\int dS \ E_T^2(\vec{b}, \omega)/(\pi) = \omega \ n(\omega)$.

We are interested by $n(\vec{b}, \omega)$ the number of equivalent photons corresponding to the energy flux through an infinitesimal surface dS of the transverse plane instead of the full transverse plane. This means that: $E_T^2(\vec{b}, \omega)/(\pi) = \omega \ n(\vec{b}, \omega)$.

- In summary: $n(\vec{b}, \omega) = \frac{1}{\pi\omega} \left| \vec{E}_T(\vec{b}, \omega) \right|^2$.
- We already know the expression of $\vec{E}_T$ in the coordinates $(\vec{b}, t)$: $\vec{E}_T(\vec{b}, t) = \frac{q}{\gamma^2} \frac{\vec{b}}{(t^2 + \frac{b^2}{\gamma^2})^{3/2}}$.
- The next step is thus to convert $\vec{E}_T(\vec{b}, t)$ in $\vec{E}_T(\vec{b}, \omega)$ after a Fourier transform.

Transform of the EM field (of equivalent photons)

- We can write:

$$\vec{E}_T(\vec{b}, \omega) = \frac{q}{\gamma^2} \int \frac{dt}{2\pi} e^{i\omega t} \frac{\vec{b}}{(t^2 + \frac{b^2}{\gamma^2})^{3/2}}.$$

Note that this kind of integral is known as a Bessel function K_1 . More precisely, $\vec{E}_T(\vec{b}, \omega)$ can be written as: $\vec{E}_T(\vec{b}, \omega) = \frac{q}{2\pi} \vec{b} \frac{\omega}{b\gamma} K_1(b\omega/\gamma)$, with the function $z \rightarrow K(z)$ which is close to a decreasing exponential.

- This is not obvious at this stage but it will be more practical to transform this integral into an integral over the transverse momentum.

For this, we use a mathematical identity, which allows to rewrite the above expression as:

$$\vec{E}_T(\vec{b}, \omega) = q \int \frac{d^2\vec{k}_T}{(2\pi)^2} e^{-i\vec{b}\vec{k}_T} \frac{(-i\vec{k}_T)}{k_T^2 + \frac{\omega^2}{\gamma^2}}$$

where $\vec{k}_T$ is the transverse momentum of photons.

Remark

In summary:

$$\vec{E}_T(\vec{b}, \omega) = q \int \frac{d^2 \vec{k}_T}{(2\pi)^2} e^{-i\vec{b}\vec{k}_T} \frac{(-i\vec{k}_T)}{k_T^2 + \frac{\omega^2}{\gamma^2}}$$

In the following, we convert this expression in real notations (to avoid some possible confusions):

$$\vec{E}_T(\vec{b}, \omega) = q \int \frac{d^2 \vec{k}_T}{(2\pi)^2} \sin(\vec{b}\vec{k}_T) \frac{\vec{k}_T}{k_T^2 + \frac{\omega^2}{\gamma^2}}$$

We have already shown that this EM field is non negligible under the condition: $\delta t < \frac{R_p}{\gamma c}$. This has several consequences:

- According to the uncertainty principle, this implies directly: $\omega < \frac{\gamma c}{R_p}$.
- Similarly, the uncertainty principle in the transverse plane implies: $k_T < \frac{c}{R_p}$. The factor γ is not present.
- The photon quadri-momentum is equal to: $k = (\omega, \vec{k}_T, \omega/v_z)$. Its square k^2 is the difference of 2 large quantities and turns out to be: $k^2 = -k_T^2 - \frac{\omega^2}{\gamma^2}$. It is not exactly zero. However, following the previous 2 conditions, the (equivalent) photon virtuality $Q^2 = -k^2 = k_T^2 + \frac{\omega^2}{\gamma^2}$ must be smaller than $\frac{c^2}{R_p^2}$. This is a coherence condition, which limits the photon virtuality to very small values (close to zero at the LHC)...

Note: posing directly the photon virtuality in the expressions of the fields is one interest of using the transverse momentum coordinates.

Equivalent photon number (2)

- Finally, in transverse momentum coordinates, the equivalent photon number (derived directly from the Poynting vector) reads: $n(\vec{b}, \omega) = \frac{q^2}{\pi\omega} \left| \int \frac{d^2\vec{k}_T}{(2\pi)^2} \sin(\vec{b}\vec{k}_T) \vec{k}_T / Q^2 \right|^2$
with $Q^2 = -k^2 = k_T^2 + \frac{\omega^2}{\gamma^2}$ the virtuality of the photons (smaller than $\frac{c^2}{R_p^2}$).
Let us remind that this is for a point-like charge q (moving at $v_z \simeq c$).
- We can generalize the previous formula to a distribution of charge (of total charge q), with the electric Form Factor (FF) $G_E(.)$:

$$n(\vec{b}, \omega) = \frac{q^2}{\pi\omega} \left| \int \frac{d^2\vec{k}_T}{(2\pi)^2} \sin(\vec{b}\vec{k}_T) \vec{k}_T G_E(k_T^2 + \frac{\omega^2}{\gamma^2}) / Q^2 \right|^2 .$$

Interestingly, $G_E(\cdot)$ is characterized by the property, that it is about 1 up to values of Q^2 of $\frac{c^2}{R_p^2}$ and quickly falls off to zero for larger values of Q^2 . This is in perfect agreement with the coherence conditions discussed above, with $\omega_{max} \simeq \gamma \frac{c}{R_p}$.

Next, we will add the contribution of the magnetic Form Factor.

Later, we will come back on a non-so-simple question: what $G_E(\cdot)$ means at the LHC?

Close to final $n(b, \omega)$

- Let us develop

$$n(\vec{b}, \omega) = \frac{q^2}{\pi\omega} \left| \int \frac{d^2\vec{k}_T}{(2\pi)^2} \sin(\vec{b}\vec{k}_T) \vec{k}_T G_E(k_T^2 + \frac{\omega^2}{\gamma^2})/Q^2 \right|^2$$

using $d^2\vec{k}_T = k_T dk_T d\phi$ and $\vec{k}_T = (k_T \cos(\phi), k_T \sin(\phi))$.

With a standard Bessel function $J_1(a) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i\phi} e^{-ia \sin \phi} d\phi$, after some manipulations:

$$n(b, \omega) = \frac{\alpha_{em}}{\pi^2\omega} \left| \int dk_T k_T^2 \frac{G_E(Q^2)}{Q^2} J_1(bk_T) \right|^2$$

as always with $Q^2 = -k^2 = k_T^2 + \frac{\omega^2}{\gamma^2}$.

- This function depends on the modulus of b (not on its direction). Also, in the last expression, the function $J_1(\cdot)$ is appearing (not $J_0(\cdot)$).

Final $n(b, \omega)$

Full expression with the magnetic FF included (using standard notations):

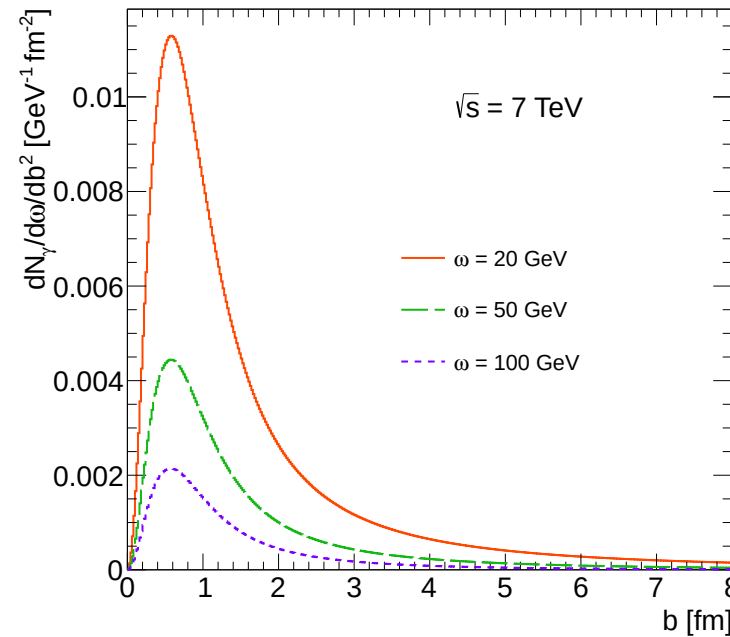
$$n(b, \omega) = \frac{\alpha_{em}}{\pi^2 \omega} \left| \int dk_T k_T^2 \frac{G_E(Q^2)}{Q^2} \left[(1-x) \frac{4m_p^2 + Q^2 \mu_p^2}{4m_p^2 + Q^2} + \frac{1}{2} x^2 \frac{Q^2}{k_T^2} \mu_p^2 \right]^{\frac{1}{2}} J_1(bk_T) \right|^2$$

where x is the energy fraction of the proton carried by the photon ($x = 2\omega/\sqrt{s}$ at the LHC).

Let us note that the electromagnetic coupling strength α_{em} is taken to be at zero virtuality (according to what we have derived), so $\alpha_{em} = 1/137.036$.

So, now: what does this distribution look like? and is it really useful to worry about the impact parameter dependence?

$n(b, \omega)$ for different energies of the photon, as function of the transverse distance b [arXiv:1410.2983]



The overall shapes of these distributions can be understood easily. At very large b values, $n(b, .)$ behaves asymptotically as $\frac{1}{b} e^{-2\omega b/\gamma}$. At very small b values, the photon distributions are damped due to the effects of the finite size of the proton (see previous slides).

Clearly, the last figure shows that $n(b, \omega)$ contains more information (physics) than the more simple standard expression for equivalent photon number $f(\omega)$. The link between both is the following:

$$f(\omega) = \int_{b>0} db \, n(b, \omega) = \frac{q^2}{\pi\omega} \int \frac{d^2\vec{k}_T}{(2\pi)^2} \left(\frac{G_E(k_T^2 + \frac{\omega^2}{\gamma^2})}{k_T^2 + \frac{\omega^2}{\gamma^2}} \right)^2 |\vec{k}_T|^2,$$

To prove the above relation, this is better to re-write $J_1(.)$ in its integral form and then, the calculation is immediate.

We need to be careful because in standard MC, only $f(\omega)$ is considered. We will come back later on this issue.

Experimental visibility

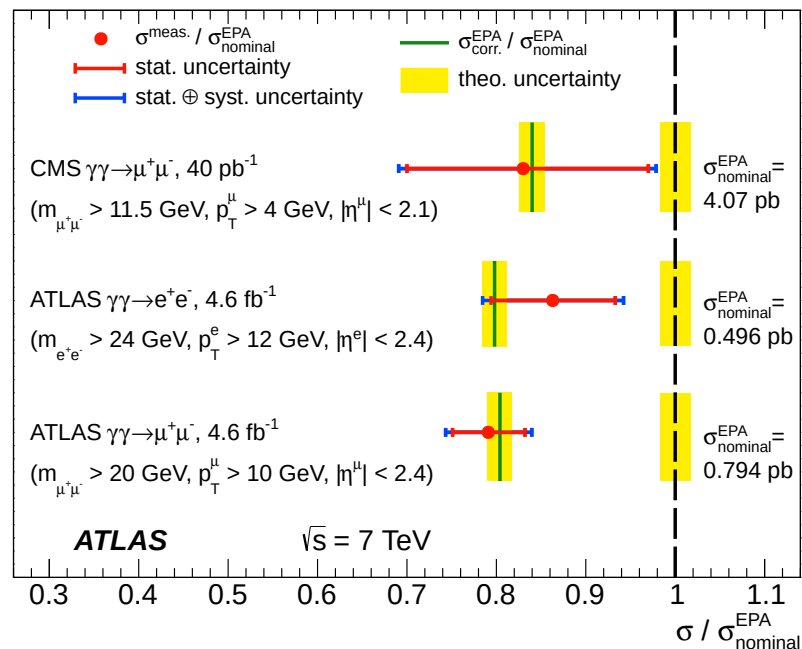
Reactions at the LHC (in ATLAS at 7 TeV): $p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + l^+l^-$.

We do not discuss how this is possible to make this measurement. This will be discussed later. This has been done [arXiv:1506.07098] and below are the results.

- $\sigma_{\gamma\gamma \rightarrow e^+e^-}^{\text{excl.}} = 0.428 \pm 0.035(\text{stat.}) \pm 0.018(\text{syst.}) \text{ pb}$,
with invariant mass of the electron pairs greater than $24 \text{ GeV}/c^2$, in which both electrons have transverse momentum $p_T > 12 \text{ GeV}/c$ and pseudo-rapidity $|\eta| < 2.4$.
- $\sigma_{\gamma\gamma \rightarrow \mu^+\mu^-}^{\text{excl.}} = 0.628 \pm 0.032(\text{stat.}) \pm 0.021(\text{syst.}) \text{ pb}$,
with invariant mass greater than $20 \text{ GeV}/c^2$, muon transverse momentum $p_T > 10 \text{ GeV}/c$ and pseudo-rapidity $|\eta| < 2.4$.

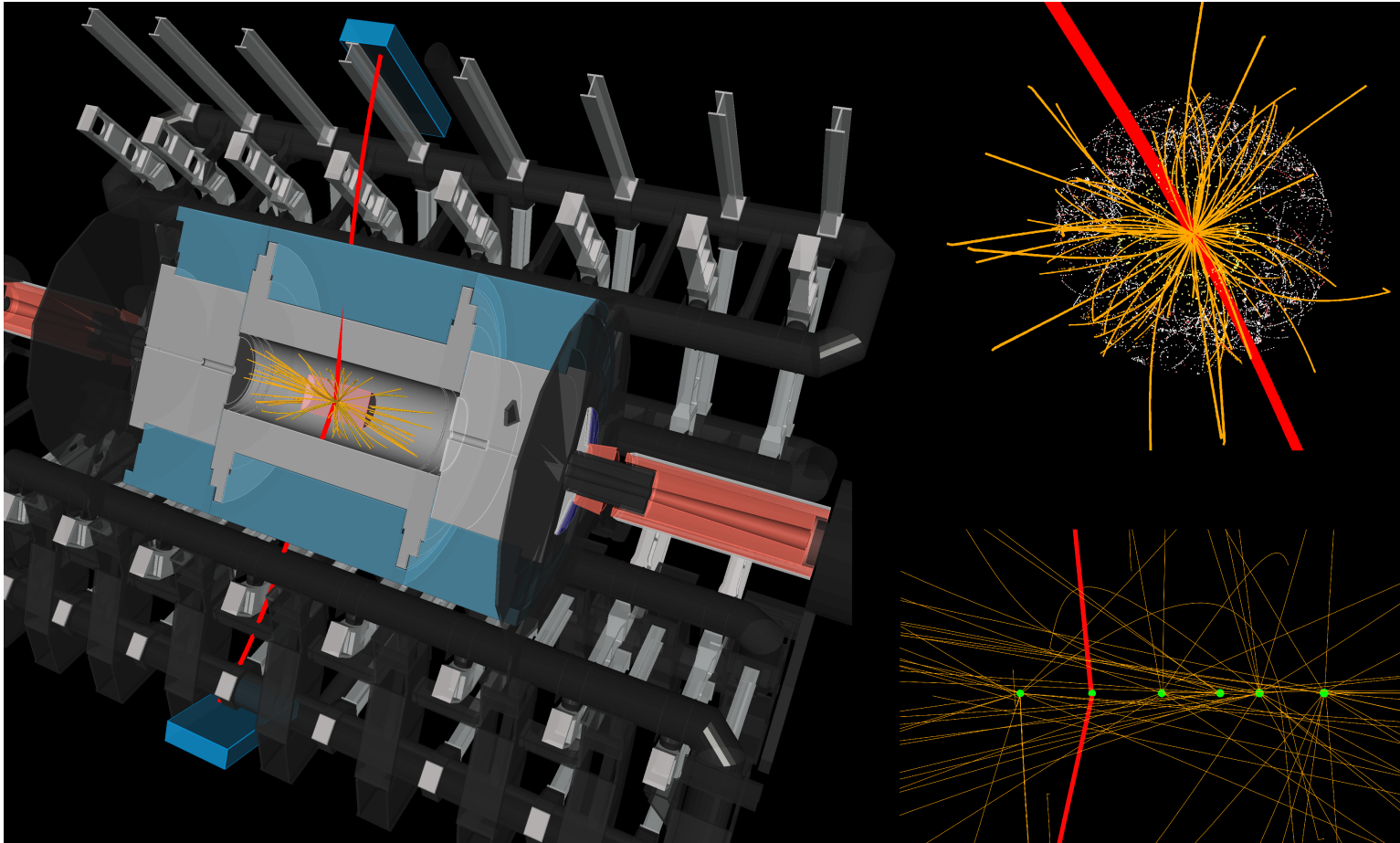
Conclusion: this is clearly visible. This justifies all previous discussion and that we can continue...

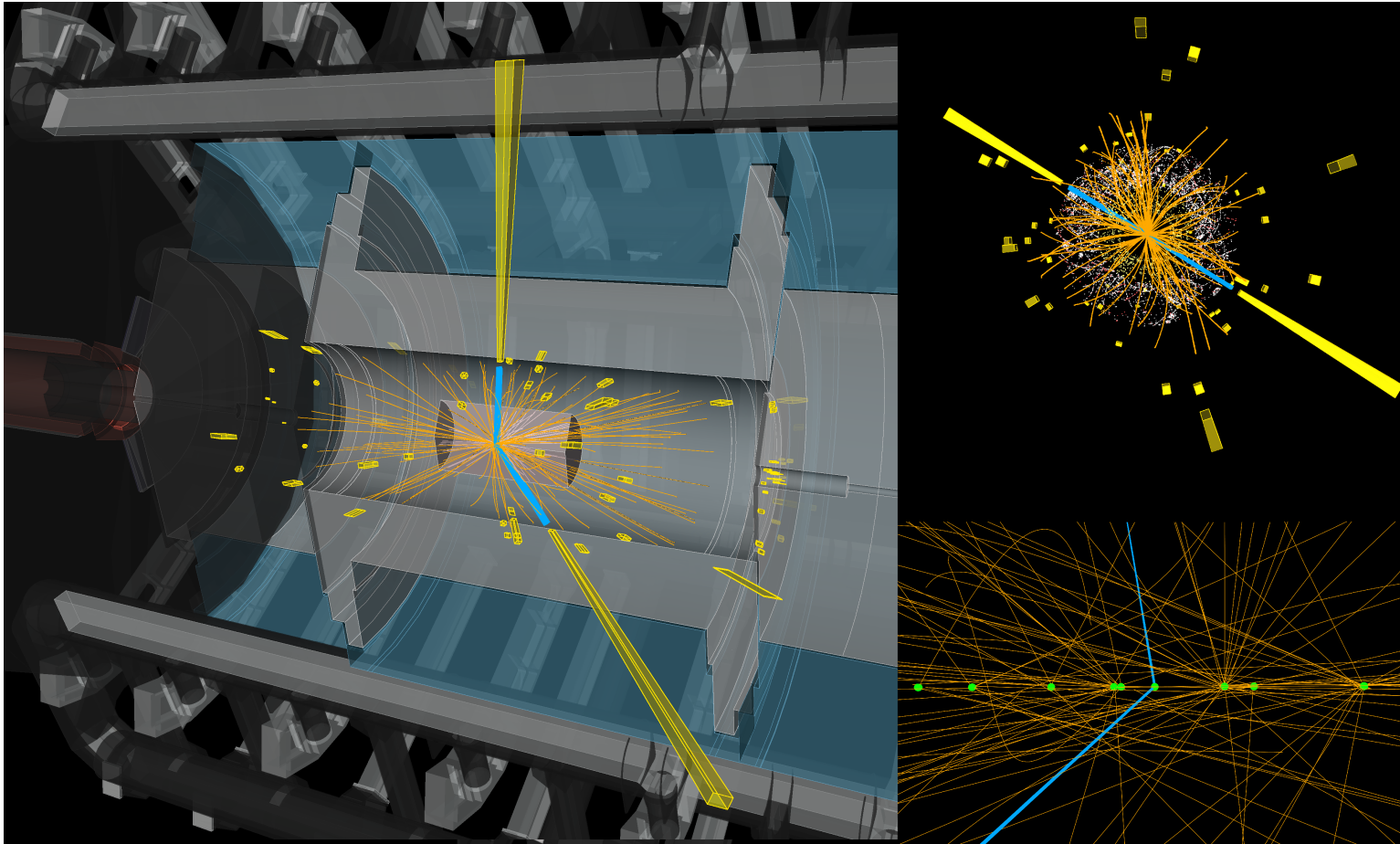
First comparisons with the theory [arXiv:1506.07098 and arXiv:1410.2983]:



EPA stands for Equivalent Photon Approximation. EPA cross sections are computed according to the initial formula: $\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + X) = \int \int f(\omega_1)f(\omega_2)\sigma_{\gamma\gamma \rightarrow X}(\omega_1, \omega_2)\frac{d\omega_1}{\omega_1}\frac{d\omega_2}{\omega_2}$, with no account of the proton finite size effects (and consequently no condition on impact parameters).

We will come back later on this plot and all that can be said on it. At this step, it shows that EPA (or corrected EPA) is not that bad. This gives the idea that the factorization (above) is reasonable. This will be our next target: prove this from fundamental arguments.





Factorization of $\gamma\gamma$ interactions at the LHC

- The general principle follows the one from standard QCD interactions leading for example to the Drell-Yan cross section...
- Here, we need to prove that this is possible to factorize the (equivalent) photon number (corresponding to the long range functions) from the $\gamma\gamma$ microscopic cross section (corresponding to the short range function).
- In order to obtain something like:
$$\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + X) = \int \int f(\omega_1) f(\omega_2) \sigma_{\gamma\gamma \rightarrow X}(\omega_1, \omega_2) \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}.$$
Or the most general form with impact parameter dependences.
- What looks like (a priori) tricky is that the last formula is a mixture of classical functions ($f(\omega)$) with QED calculations (the microscopic $\gamma\gamma$ cross section).

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- The general expression (in pure QED) for producing a pair of fermions or bosons (written generically as L below) via a photon-photon interaction at the LHC,
 $\sigma(p + p \rightarrow pp(\gamma(k_1)\gamma(k_2)) \rightarrow p + p + L^-(k_3)L^+(k_4))$ reads:

$$\int d^2\vec{b} \int d\Gamma_{(3,4)} \left| \int d\Gamma_{(1,2)} A_1^\mu(k_1, \vec{b}) T_{\mu\nu}(k_1, k_2, k_3, k_4) A_2^\nu(k_2) \right|^2.$$

where the momenta of particles are labeled (k_1, k_2, k_3, k_4) (see above) and the phase space integrands are written as $d\Gamma_{(1,2)}$ and $d\Gamma_{(3,4)}$. For the simplicity of notations, we have included the conservation law $(2\pi)^4 \delta^4(k_1 + k_2 - k_3 - k_4)$ into the transition amplitude $T_{\mu\nu}(k_1, k_2, k_3, k_4)$.

- Obviously, the functions $A_1^\mu(k_1, \vec{b})$ and $A_2^\nu(k_2)$ are the EM potentials produced by the colliding protons, whose velocities are $u_1 = \gamma(1, v, 0, 0)$ and $u_2 = \gamma(1, -v, 0, 0)$, with the condition that the impact parameter of the reaction is b .
- The amplitude $T_{\mu\nu}(k_1, k_2, k_3, k_4)$ is calculable from QED rules for the reaction:
 $\gamma(k_1)\gamma(k_2) \rightarrow L^-(k_3)L^+(k_4)$.

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- Then, the architecture of the proof of factorization is clear:
 - (1) $T_{\mu\nu}(k_1, k_2, k_3, k_4)$ is known from any book on QED,
 - (2) $A_1^\mu(k_1, \vec{b})$ and $A_2^\nu(k_2)$ are also known (see the first slides),
 - (3) consequently, we can compute the big integral of the previous slide and
 - (4) check that $\sigma(p + p \rightarrow pp(\gamma(k_1)\gamma(k_2)) \rightarrow p + p + L^-(k_3)L^+(k_4))$ can be expressed under its factorized form.
 - We will not do it. It works!

Here, we will give the an original trick that gives most of the idea of the proof.

- A key point is to realize (in a way or another) that $A_1^\mu(k_1, \vec{b})T_{\mu\nu}(k_1, k_2, k_3, k_4)A_2^\nu(k_2)$ is zero for exactly real photons ($k_1^2 = 0$ and $k_2^2 = 0$).
Then, we recall that in our physics case, we are not dealing with exactly real photons, but we have shown that: $k^2 = -k_T^2 - \frac{\omega^2}{\gamma^2}$, where k_T and $\frac{\omega}{\gamma}$ are small parameters. Indeed, $\mathcal{O}(k_T) = \mathcal{O}(\frac{\omega}{\gamma}) = \mathcal{O}(\omega)/\gamma = \mathcal{O}(k_{//})/\gamma$.
- Then, for almost real photons, $A_1^\mu(k_1, \vec{b})T_{\mu\nu}(k_1, k_2, k_3, k_4)A_2^\nu(k_2)$ will not be zero but proportional to the small parameters of the calculations, thus proportional to terms like $\vec{k}_{T,1} \cdot \vec{k}_{T,2}$.
- After taking the square of this quantity, this will give terms of orders $\left| \vec{k}_{T,1} \cdot \vec{k}_{T,2} \right|^2$. This means that there will be an integration of terms in $\left| \vec{E}_{T,1}(\omega_1, \vec{b}_1) \cdot \vec{E}_{T,2}(\omega_2, \vec{b}_2) \right|^2$ and these are exactly the (equivalent) photon numbers.

Charge q moving at constant $v = v_z \simeq c$ alternative derivation of the equivalent EM field

As we have already seen (in first slides), this is interesting to find directly the -equivalent- EM field in terms of $\vec{k}_T$. For this purpose, the most natural solution is to solve directly the field equation: $\partial_\mu \partial^\mu A^\nu(x) = j^\nu(x)$, where $j^\nu(x)$ describes the behavior of the proton and $A^\nu(x)$ will be the -equivalent- EM field (associated to this current).

Proposition 3. *The -equivalent- EM field reads: $A^\nu(k) = -\frac{1}{k^2} j^\nu(k) = -2\pi \delta(k.u) \frac{\rho(\sqrt{-k^2})}{k^2} u^\nu = r(k) u^\nu$ where $u^\nu = \gamma(1, 0, 0, v_z)$ is the four-velocity of the proton. Interestingly, $r(k)$ is built with Lorentz invariant quantities.*

Proof. In the proton rest frame, we can write: $j'^\nu(x') = \rho(|x'|) u'^\nu$ where $u'^\nu = (1, 0, 0, 0)$. Then, this is obvious to transform $j'^\nu(x')$ in $j'^\nu(k')$, where $k'^\nu = (\omega', \vec{k}')$.

It reads: $j'^\nu(k') = \int d^4x' e^{ik'x'} j'^\nu(x') = 2\pi \delta(\omega') \rho(|k'|) u'^\nu$. The key observation is then that this can be rewritten in a covariant way as: $j'^\nu(k') = 2\pi \delta(k'.u') \rho(\sqrt{-k'^2}) u'^\nu$, which concludes the proof. $\square$

When we consider the electric Form Factor $G_E(-k^2)$, $r(k)$ becomes immediately: $-2\pi \delta(k.u) \frac{q G_E(\sqrt{-k^2})}{k^2}$. As before, we have: $k^2 = -k_T^2 - \frac{\omega^2}{\gamma^2}$.

Also, $F^{\mu\nu} = -i(k^\mu A^\nu(k) - k^\nu A^\mu(k))$ which gives: $\vec{E}_T(k) = -i A^0(k) \vec{k}_T$, which concludes.

What we have done so far

(1) We have derived the -equivalent- photon spectrum for a charge q moving at constant $v = v_z \simeq c$ (protons or ions at the LHC). This gives the so-called equivalent photon numbers $n(\vec{b}, \omega)$, defined from the conservation of energy (using the Poynting vector).

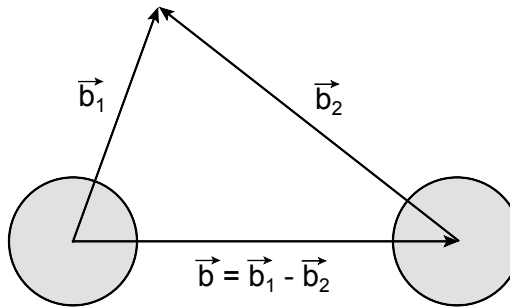
(2) Then, at the LHC, a reaction is possible mediated via the photon-photon interaction. There are some conditions that guarantee the existence of this reaction (what we have called coherence conditions): $b_{1,2} > R_p$ and $\omega_{max} = \frac{\gamma}{R_p}$. As usual with, $k^2 = -k_T^2 - \frac{\omega^2}{\gamma^2}$ and $\mathcal{O}(k_T) = \mathcal{O}(\frac{\omega}{\gamma}) = \mathcal{O}(\omega)/\gamma$. Under these conditions, we have shown that this cross section (in pure QED) can then be written as a factorized expression of the photon numbers times the microscopic photon-photon:

$$\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + X) = \int n(\vec{b}_1, \omega_1) n(\vec{b}_2, \omega_2) \sigma_{\gamma\gamma \rightarrow X}(\omega_1, \omega_2) \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2} d^2\vec{b}_1 d^2\vec{b}_2$$

where only one integration symbol is posed (for simplicity of the notations)...

There is still one point to consider

(3) If we had written completely the factorization proof, we would have found another conditions on the impact parameters. See figure below, where we present a schematic view of the two protons and the transverse distances $\vec{b}_1$ and $\vec{b}_2$. This is clear from this view that the geometrical non-overlapping condition of the two protons corresponds to $|\vec{b}_1 - \vec{b}_2| > 2R_p$.



This is the last condition to be considered. It looks like trivial but it makes a correlation between the 2 impact parameters, which complexifies the integral of the previous slide...

There is another way to encode this non-overlapping condition by computing the probability of no inelastic interactions between the 2 colliding protons. Then, we replace the integration condition $|\vec{b}_1 - \vec{b}_2| > 2R_p$ by a new function ($P_{non-inel}(|\vec{b}_1 - \vec{b}_2|)$) factorized in the big integral. We have done this: arXiv:1410.2983. As both methods give the same results, we will continue with the simplest geometrical one in the following ($|\vec{b}_1 - \vec{b}_2| > 2R_p$).

Comparing to MC predictions

As already mentioned, existing MCs do not consider the impact parameter dependences and related conditions. Thus, MCs provide:

$$\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + X) = \int f(\omega_1) f(\omega_2) \sigma_{\gamma\gamma \rightarrow X}(\omega_1, \omega_2) \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}$$

In order to correct this prediction to what we have shown to be reasonable (at this point), we need to multiply this MC calculation by a factor:

$$S_{\gamma\gamma}^2 = \frac{\int_{b_{1,2} > R_p, |\vec{b}_1 - \vec{b}_2| > 2R_p} n(\vec{b}_1, \omega_1) n(\vec{b}_2, \omega_2) d^2\vec{b}_1 d^2\vec{b}_2}{\int_{b_{1,2} > 0} n(\vec{b}_1, \omega_1) n(\vec{b}_2, \omega_2) d^2\vec{b}_1 d^2\vec{b}_2}$$

We have called $S_{\gamma\gamma}^2$ the suppression factor. This is a function of all parameters of the collision.

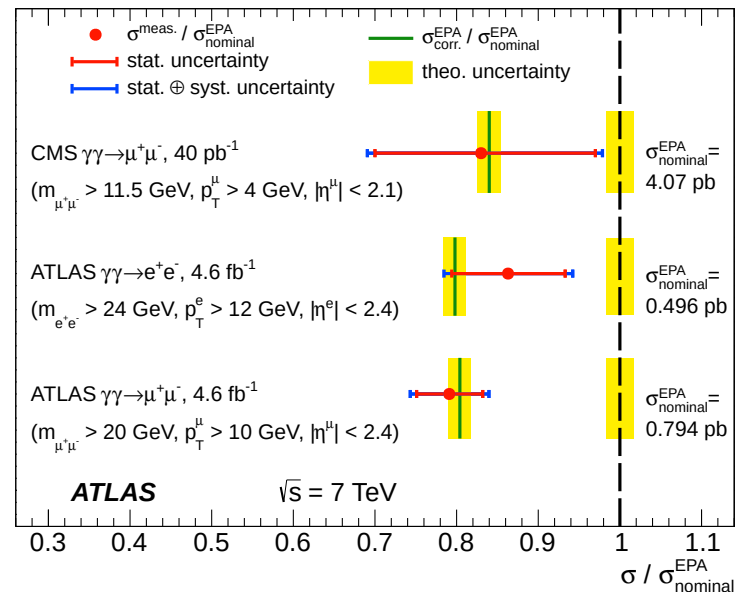
$S_{\gamma\gamma}^2$ in practice

Comparison of total cross sections at $\sqrt{s} = 13$ TeV for different processes $pp(\gamma\gamma) \rightarrow ppX$ with and without suppression factor applied [arXiv:1410.2983].

Process	σ_{tot}	$\sigma_{tot} \otimes S_{\gamma\gamma}^2$	$\langle S_{\gamma\gamma}^2 \rangle$
$\gamma\gamma \rightarrow H$ ($M_H = 125$ GeV)	0.15 fb	0.11 fb	0.74
$\gamma\gamma \rightarrow \mu^+\mu^-$ ($W_{\gamma\gamma} > 40$ GeV)	12 pb	10 pb	0.8
$\gamma\gamma \rightarrow \mu^+\mu^-$ ($W_{\gamma\gamma} > 160$ GeV)	36 fb	25 fb	0.7
$\gamma\gamma \rightarrow W^+W^-$	82 fb	53 fb	0.65
$\gamma\gamma \rightarrow \gamma\gamma$ ($W_{\gamma\gamma} > 200$ GeV)	0.06 fb	0.04 fb	0.64

So, the order of the effect is a reduction (suppression) between 20 % to 40 % of the predicted cross section. We can not miss this if we make a proper measurement.

It is time to come back on the experimental results



EPA is the (nominal) MC prediction. The prediction with the vertical green line is what we compute from a full treatment of the impact parameter effects, that originate from the finite size of the protons.

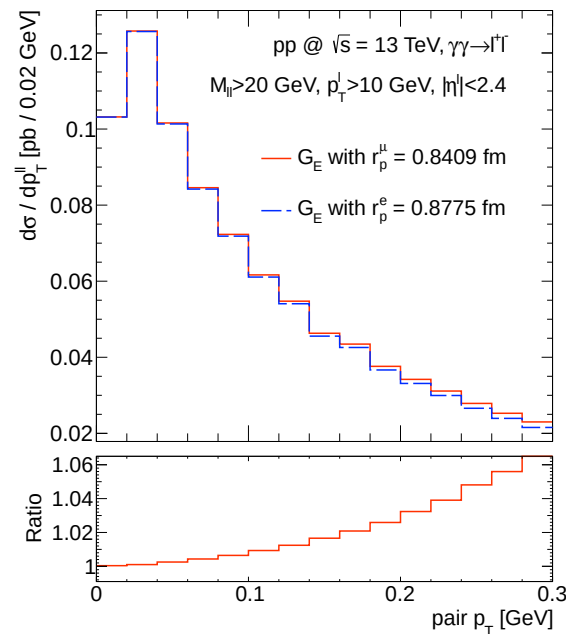
We see that we have very well understood the data. It was the first time it was done. At this point, we can state that: (1) this is a great success to prove that this is possible to measure such processes at the LHC with a good precision (see error bars) and (2) this is also a great success to understand precisely the experimental results...

Something more

As it can be seen on the previous figure, we observe a difference between electrons and muons.

This is not the first time this effect is measured since 1960/70 (and always in the same sense).

At present level of precision, we can not really move further but **there is certainly a path for testing the lepton universality in exclusive di-lepton reactions at the LHC (with a much higher precision).**



Chapter 2

Inelastic (protons) scattering

$$p + p \rightarrow pp(\gamma\gamma) \rightarrow p^{(*)} + p + X \text{ (Single Dissociation)}$$

$$p + p \rightarrow pp(\gamma\gamma) \rightarrow p^{(*)} + p^{(*)} + X \text{ (Double Dissociation)}$$

Methodology

The key idea is to come back to the so-called hadronic tensor $W^{\mu\nu}$, which can be expressed in terms of 2 independent functions $W_{1,2}(\nu, k^2)$ of 2 variables ν and k^2 (with $(p+k)^2 = m_p^2 + 2m_p\nu + k^2$).

Let us remind the expressions of these functions in the elastic physics case (see chapter I):

- (1) $W_{1,2}(\nu, k^2) = \tilde{W}_{1,2}(k^2)\delta(\nu + \frac{k^2}{2m_p})$, where the $\delta(\cdot)$ function translates the elastic requirement: $(p+k)^2 = m_p^2 + k^2 + 2m_p\nu = m_p^2$. Also, we remind the relation: $k^2 = -k_T^2 - \frac{\omega^2}{\gamma^2}$.
- (2) With $\tilde{W}_2 = q^2(G_E^2(\cdot) + \tau G_M^2(\cdot))/(1 + \tau) \simeq q^2|G_E(-k^2)|^2$ and $\tilde{W}_1 = q^2 \frac{-k^2}{4m_p^2} |G_M(-k^2)|^2 \ll \tilde{W}_2$ (with $\tau = \frac{-k^2}{4m_p^2} \ll 1$). We keep the charge q outside the functions $G(\cdot)$ as in previous slides.

Therefore, we need to find the equivalent to (1) and (2) but this time for the inelastic case, when $(p+k)^2 = m_p^2 + k^2 + 2m_p\nu \neq m_p^2$.

We need to find the inelastic functions $W_{1,2}(\nu, k^2)$. As the variable x has already been used (chapter I), we take the notation $z = \frac{-k^2}{2m_p\nu}$ (equal to 1 for the elastic case and < 1 for inelastic reactions).

We can distinguish 2 configurations:

- At low excitation energies, discrete states will dominate (resonances of the proton). These resonances will be at well defined discrete energies (neglecting their finite width), and ν has to be equal to Δ , where Δ is the energy of the excited state in its rest frame.

We can write: $(p - k)^2 = (m_p + \Delta)^2$ and $W_{1,2}(\nu, k^2) = \tilde{W}_{1,2}(k^2)\delta(\nu - \Delta)$.

Neglecting terms quadratic, we obtain also: $k^2 = -k_T^2 - \frac{\omega^2}{\gamma^2} - 2\frac{\omega\Delta}{\gamma} - \frac{\Delta^2}{\gamma^2}$.

Obviously, as long as: $\omega > \gamma\Delta$, we can neglect the terms in Δ in this relation.

- For higher excitation energies, we know the fundamental relation: $\nu W_2(\nu, k^2) = F_2(z)$. In the following, we make the assumption that all inelastic configurations can be encoded into this last expression.

The next step is immediate: the inelastic cross sections can be obtained by making the replacement: $q^2|G_E(-k^2)|^2 \rightarrow \frac{F_2(z)}{z}$ and then integrating on $\int dz/z$. Something similar can be done magnetic Form Factor... The full formulae are complicated but the principle is exactly what is described here. Of course, in order to get a realistic estimate of the inelastic cross sections (single or double dissociations), $F_2(z)$ must be computed from a QCD evolution included QED effect (and the distribution function associated to the photon structure of the proton -see later-).

Following our previous statements, we can always write the inelastic photon number (function of the fraction of the photon energy (x) and the scale Q^2) as:

$$\frac{d^2 f_{inel}(x, Q^2)}{dx d \log Q^2} \sim \frac{\alpha_{em}}{2\pi} \int_x^1 \frac{dz}{z} P_{\gamma \leftarrow q}(x/z) \frac{F_2(z, Q^2)}{z} K(z, x)$$

where $P_{\gamma \leftarrow q}(y) = \frac{1+(1-y)^2}{y}$ is the splitting function that appears in the evolution equation linking the the photon distribution (in the proton) and quarks, and where $K(z, x)$ is a kinematic factor.

QCD-QED evolution equations (remarks)

QCD evolution equations of PDF is a well established technique (since long). This is useful to remember than LO QED effects can be comparable to NLO QCD corrections as:

$$\mathcal{O}(\alpha_s^2)/\mathcal{O}(\alpha_{em}) \sim (\alpha_s^2)(M_Z^2)/(\alpha_{em})(M_Z^2) \sim 0.11^2/(1/127) \simeq 1.8.$$

Therefore, at NLO (in QCD) this is essential to introduce something like a photon DF (distribution function) that will play the same role as quarks and gluons in the equations, all linked by splitting functions.

For example, the momentum sum rule will change as the photons will carry part of the momentum!

Inelastic (photon-photon) cross sections at the LHC

Finally, it is possible to get the inelastic (equivalent) photon number $f_{inel}(\cdot)$ such that we can predict:

(a) Single dissociation:

$$\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p(*) + p + X) = S_{\gamma\gamma}^2 \cdot \int f_{inel}(\omega_1) f(\omega_2) \sigma_{\gamma\gamma \rightarrow X}(\omega_1, \omega_2) \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}$$

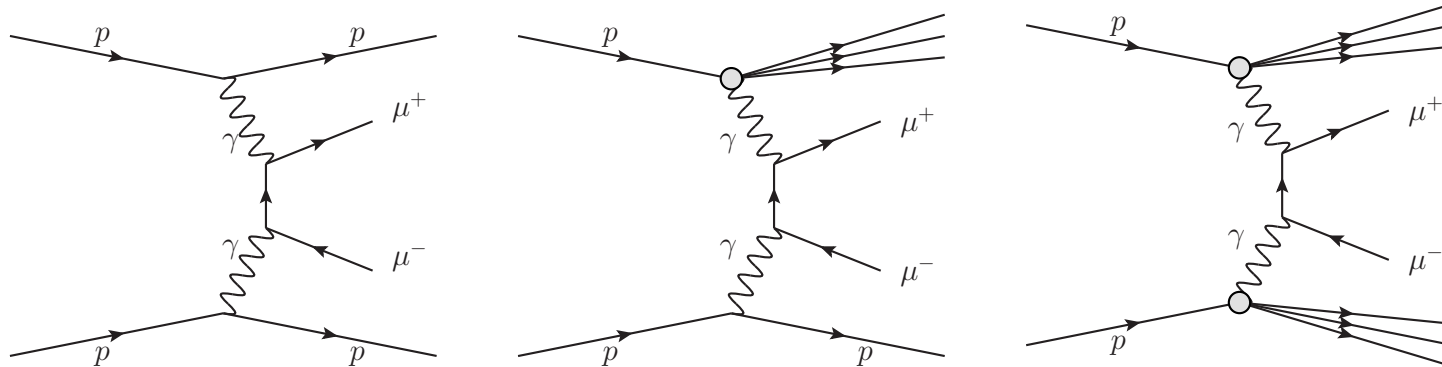
(b) Double dissociation:

$$\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p(*) + p(*) + X) = S_{\gamma\gamma}^2 \cdot \int f_{inel}(\omega_1) f_{inel}(\omega_2) \sigma_{\gamma\gamma \rightarrow X}(\omega_1, \omega_2) \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}$$

There is no essential reason why $S_{\gamma\gamma}^2$ should be the same for elastic and inelastic cross sections... The problem is that we do not know how to evaluate the impact parameter dependence for $f_{inel}(\cdot)$. Thus, for simplicity, we assume this factor to be the same. Moreover, we have measured it to be consistent for both processes (for $X = \ell^+ \ell^-$).

Also, because $f_{inel}(\cdot)$ contains an integration of terms in $F_2(z)/z$ where z can be very small, the inelastic cross sections can be expected to dominate completely the elastic one (in the absence of specific analysis requirements).

In pictures for the production of a pair of muons ($X = \mu^+ \mu^-$)



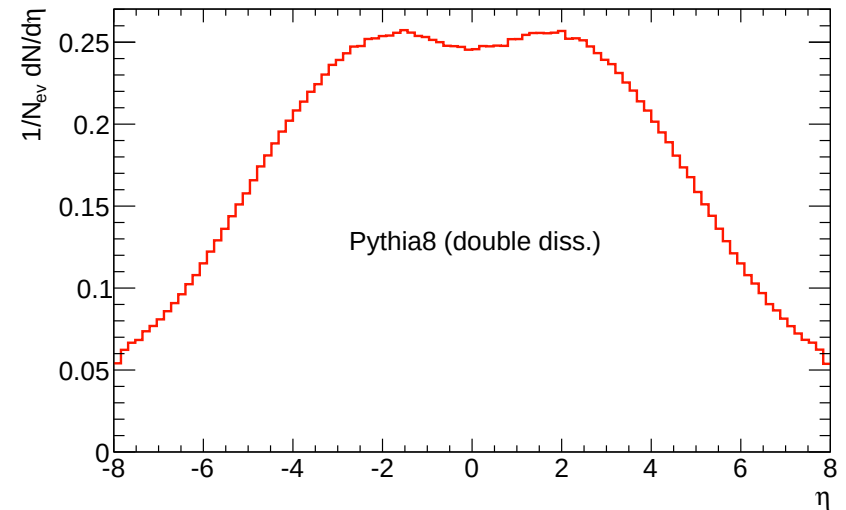
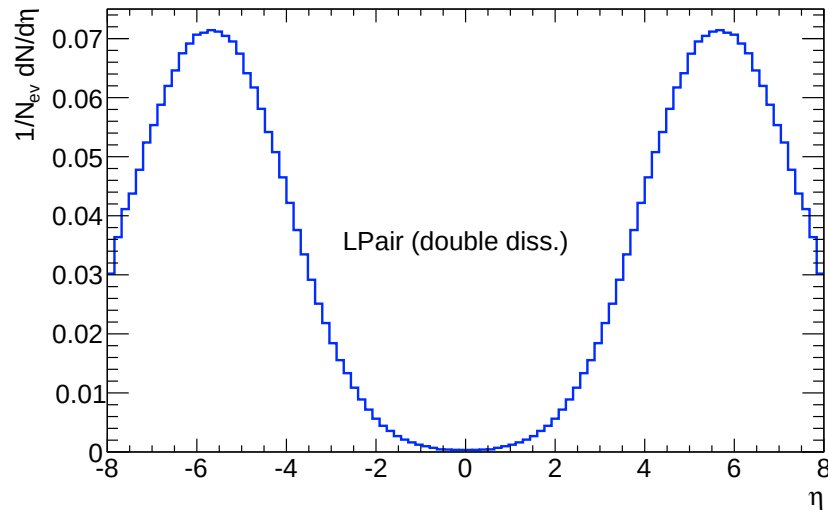
The left figure stands for the pure elastic physics case, and then Single and Double dissociations. Then, the full cross sections can be written using terms in:

$$f(\cdot)f(\cdot) + 2f_{inel}(\cdot)f(\cdot) + f_{inel}(\cdot)f_{inel}(\cdot) = [f(\cdot) + f_{inel}(\cdot)]^2$$

As a rule of thumb, the full cross section (in any given kinematic range) is about 10 times the pure elastic cross section.

-
- We have already seen that the photon-induced elastic cross section of $\mu^+\mu^-$ for $M_{\mu^+\mu^-} > 40$ GeV is $\simeq 10$ pb.
 - This means that the full photon-induced cross section for $\mu^+\mu^-$ production (elastic and inelastic) will be around 100 pb.
 - This is comparable to the standard Drell-Yan (DY) cross section in the same kinematical range!
 - Therefore, photon-induced processes are also essential for key precise (Standard Model) SM measurements at the LHC.
 - What happens in practice? When the DY analysis is done, there are a lot of experimental requirements dedicated to the control of the central hadronic activity. When this is done, photon-induced events are strongly suppressed. At the end, only a few % of the photon-induced cross section is still visible.
 - However, knowing precisely this few % is important to reach the high precision at the LHC (independently of the photon-photon physics for itself).

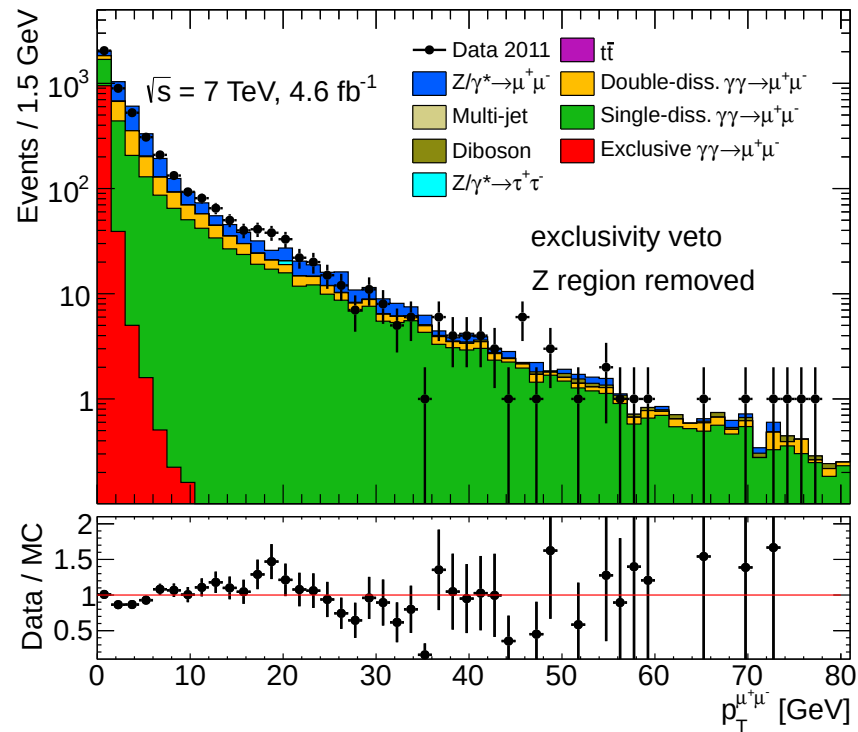
Particle production in MCs



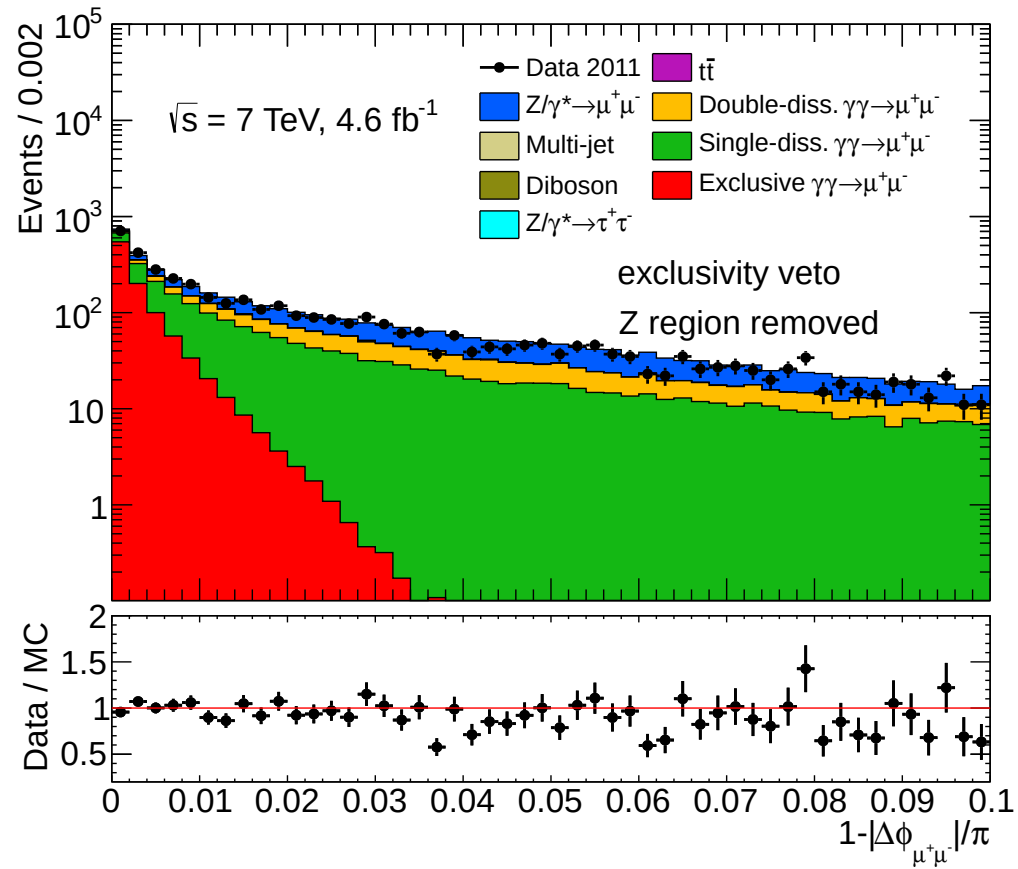
- Another important difficulty: the particle emission for the inelastic cases in MCs. Ideally, this is essential to have a proper description of this point.
- MCs do not agree. Pythia8 is the most involved (according to what we have explained in the previous slides).
- *Note however: when we suppress the central hadronic activity, we can expect that both MCs should not be too different.*

Inelastic $\gamma\gamma \rightarrow \mu^+\mu^-$ at the LHC, data and MCs

Here all the hadronic activity is suppressed in the central region ($|\eta| < 2.5$) through the experimental analysis. SD is simulated with LPair. The description of the measurement is correct. However, remind that most of the inelastic cross section has been suppressed by vetoing the central hadronic content [arXiv:1506.07098].



Similarly for the acoplanarity distribution.



Chapter 3

Form Factors

What is $G_E(\cdot)$ in: $W_2(\nu, k^2) \simeq q^2 |G_E(-k^2)|^2 \delta(\nu + \frac{k^2}{2m_p})$?

(this will be a small -but dense- chapter)

Position of the problem

- This is always possible to define functions $G(.)$ such that for example for elastic reactions:
$$W_2(\nu, k^2) \simeq q^2 |G_E(-k^2)|^2 \delta(\nu + \frac{k^2}{2m_p}).$$
- What is not-so-simple is its interpretation.
- In general a FF is defined as: $\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega}\right)_{point} |F(.)|^2$
- The problem of the interpretation is the following for relativistic systems: consider a system of size R and mass m . The system cannot be localized to a precision better than its Compton wavelength $1/m$. Any attempt to do this with an external potential will result in creation of particle-antiparticle pairs.

-
- As a consequence, the static size of the system cannot be defined to a precision better than $1/m$. If $R \gg 1/m$, which is the case for all non-relativistic systems, this is not a significant constraint. Example: the hydrogen atom for which $R.m \sim 10^5$.
 - However, when the probing wavelength is comparable to $1/m$, the form factors are no longer determined by the internal structure alone. They contain also the dynamical effects of Lorentz boosts because the initial and final protons have different momenta.

A solution

Proposition 4. *A correct statement: the FF can be interpreted as Fourier transform of a charge distribution w.r.t. impact parameter in the infinite momentum frame. **This is exactly what we need: indeed, in this case, $G_E(\cdot)$ can be related to the proton charge distribution in the transverse plane (as measured at HERA for example).***

Proof. We consider only the electric charge and neglect any magnetic effects. This means that we also forget about the spin in the wave functions formulae. Then, the FF is exactly $G_E(\cdot)$. We take $q = 1$.

By definition: $2p^+ G_E(-k_T^2) = \langle p' | j^+(\vec{0}_T) | p \rangle$. Our purpose is then to relate this quantity to the charge distribution in the transverse plane (as a function of what we have defined as the impact parameter $\vec{b}$).

By definition (again): $\rho(\vec{b}) = \frac{1}{2p^+} \langle p^+, \vec{R}_T = \vec{0}_T | j^+(\vec{b}) | p^+, \vec{R}_T = \vec{0}_T \rangle$

This last expression is less trivial (or common) as the first one. It includes the charge density operator, averaged over a state of the system, with a given momentum and position in the transverse plane (see later).

We have:

$$|p^+, \vec{R}_T = \vec{0}_T \rangle = N \int d^2 \vec{p}_T |p^+, \vec{p}_T \rangle .$$

Then, by computing $\rho(\vec{b})$, using the (Heisenberg) relation

$$\langle p^+ \vec{p}'_T | j^+(\vec{b}) | p^+, \vec{p}_T \rangle = \langle p^+ \vec{0}_T | j^+(\vec{0}) | p^+, \vec{0}_T \rangle e^{-i \vec{k}_T \vec{b}},$$

we obtain the final result that we want:

$$\rho(\vec{b}) = \int \frac{d^2 \vec{k}_T}{(2\pi)^2} G_E(-k_T^2) e^{-i \vec{k}_T \vec{b}}.$$

□

In summary, FF makes sense in the common sense only for non-relativistic systems. Then, the interpretation in terms of imaging the 3D structure of the object probed is correct. There is a 3D Fourier transform relation between the charge density (in 3D) and the FF.

For relativistic systems (spin 0 or 1/2), the situation is different. At the LHC (for example), the Fourier transform between a FF and a charge distribution is valid only in the transverse plane (or the impact parameter plane). This means in 2D.

This comes from the fact this is not possible to localize a system in 3D of space (for those systems). However, as we have shown: **a basis of orthogonal states** is a mixture of the momentum (the fast component) and the transverse spatial structure (that can be localized in this context): $|p^+, \vec{R}_T = \vec{0}_T \rangle$.

This is not limited to FF

The definition: $\rho(\vec{b}) = \frac{1}{2p^+} \langle p^+, \vec{R}_T = \vec{0}_T | j^+(\vec{b}) | p^+, \vec{R}_T = \vec{0}_T \rangle$ can be extended to the physics case of a parton distribution function (PDF) which also includes the longitudinal momentum dependence (zp^+):

$$q(z, \vec{b}) = \int \frac{dz^-}{4\pi} \langle p^+, \vec{R}_T = \vec{0}_T | \bar{\psi}(-z^-/2, \vec{b}) \gamma^+ \psi(z^-/2, \vec{b}) | p^+, \vec{R}_T = \vec{0}_T \rangle e^{izp^+z^-}$$

Note: there is no integration on the (internal) variable z^+ which is zero (exactly). Then, this is possible to use the relation: $|p^+, \vec{R}_T = \vec{0}_T \rangle = N \int d^2\vec{p}_T |p^+, \vec{p}_T \rangle$ and convert this integral into the transverse Fourier transform of a function, which will depend on z and $-\vec{k}_T^2$.

This looks like purely academic, but all this is hidden in MCs when dealing with MPI (multi-parton interactions) and more specifically in DPS (double parton scattering).

For completeness

Extend: $|p^+, \vec{R}_T = \vec{0}_T \rangle = N \int d^2 \vec{p}_T |p^+, \vec{p}_T \rangle$ to any impact parameter space position ($\vec{b}$):

$$|p^+, \vec{b} \rangle = N \int d^2 \vec{p}_T e^{-i \vec{p}_T \vec{b}} |p^+, \vec{p}_T \rangle$$

The normalization constant N turns out to be: $1/(2\pi)^2$.

All substitutions done, the formula of the last slide reads:

$$q(z, \vec{b}) = \int \frac{d^2 \vec{k}_T}{(2\pi)^2} H_q(z, 0, -k_T^2) e^{-i \vec{k}_T \vec{b}}.$$

Where $H_q(\cdot)$ is a generalized PDF (which can be measured). A very important remark comes with the observation that

$$\langle b^2 \rangle (z) = \frac{\int d^2 \vec{b} b^2 q(z, \vec{b})}{\int d^2 \vec{b} q(z, \vec{b})} = 4 \frac{\partial}{\partial t} \log H_q(z, 0, t)|_{t=0}$$

If we can measure $H_q(\cdot)$, then we can measure $\langle b^2 \rangle (z)$ (proton tomography).

Chapter 4

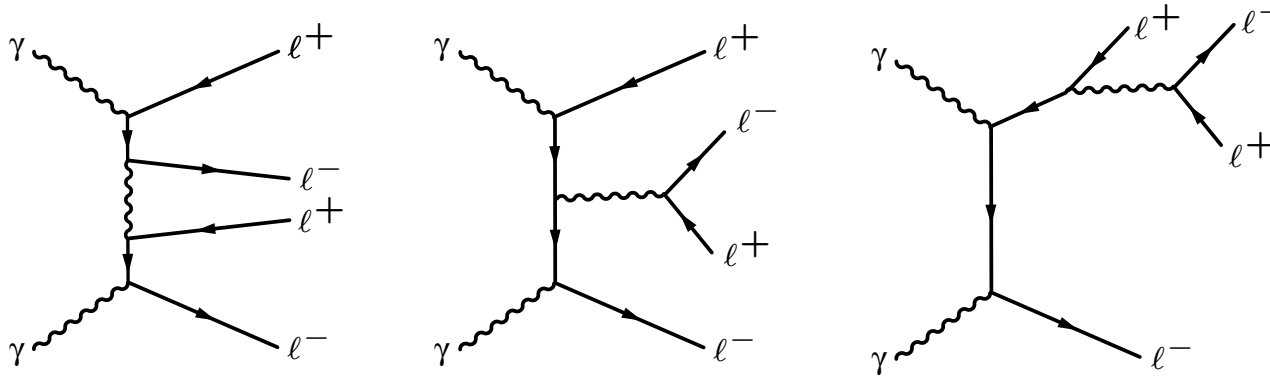
Moving to a more complex process [arXiv:1511.02065]:

$$p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + 4\ell$$

for which we will verify all our previous statements...

In pictures

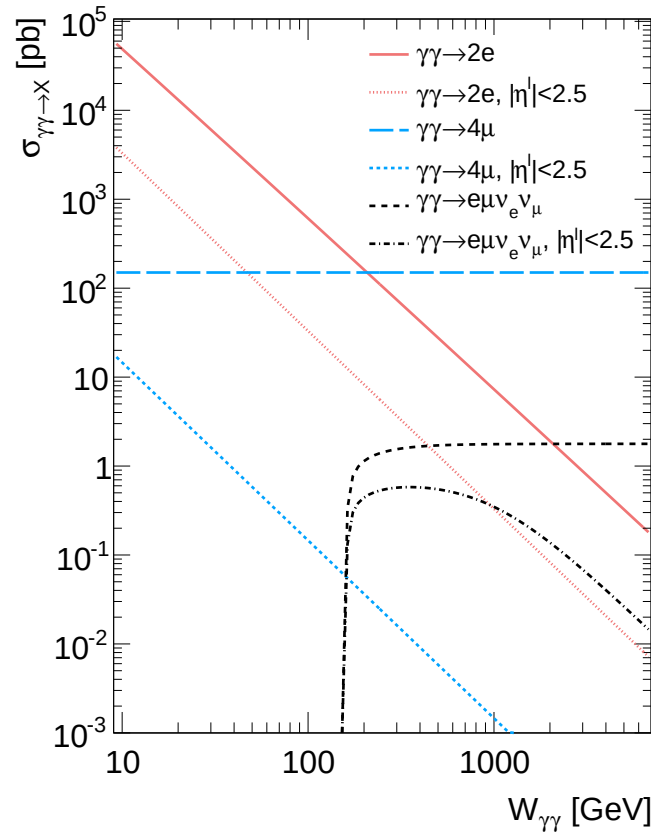
Examples of leading-order diagrams for the process $\gamma\gamma \rightarrow 4\ell$. All contributing diagrams form terms of $\mathcal{O}(\alpha_{em}^4)$, where α_{em} is the electromagnetic coupling constant.



First picture: The two photons radiate a pair of leptons and two of these leptons undergo scattering (the t-channel photon exchange) with its characteristic singularity in the forward direction (when t tends towards zero).

See next slide...

Leading-order cross sections for the elementary processes $\gamma\gamma \rightarrow X$ as a function of the energy of the two-photon system, $W_{\gamma\gamma}$. Different electroweak final states are pictured with and without requirements on the polar angles of leptons ($|\eta^\ell| < 2.5$). The cross section for lepton-neutrino pair production involves decays of W bosons.



Comments and Methodology

- Total $\gamma\gamma \rightarrow 4\ell$ and $\gamma\gamma \rightarrow W^+W^-$ cross sections are constant as a function of $W_{\gamma\gamma}$. In each case this constant behavior is due to the spin-1 t -channel particle exchange (photon or W boson respectively), which implies that the cross section is completely dominated by configurations where the leptons are emitted in very forward directions, almost collinear to the beams.
- After some angular cuts are imposed to avoid these configurations, all cross sections behaves in $1/W_{\gamma\gamma}^2$ (see figure).
- In practice, all $\gamma\gamma$ cross sections and distributions have been computed with Madgraph5 (aMC@NLO) and interfaced with Pythia8 for pp collisions [arXiv:1511.02065].
- $\rightarrow$ This makes possible a correct treatment (according to the theory exposed here) for elastic and inelastic physics cases. Obviously, this method is not limited to 4ℓ production...

$$\sigma(p + p \rightarrow pp(\gamma\gamma) \rightarrow p + p + 4\ell)$$

Total cross sections for $\gamma\gamma \rightarrow 4\ell$ ($\ell = e, \mu$) production in pp collisions for $\sqrt{s} = 8$ TeV and 13 TeV. Leptons with $|\eta^\ell| < 2.5$ and $p_T^{\ell 1} > 10$ GeV, $p_T^{\ell 2} > 10$ GeV, $p_T^{\ell 3} > 1$ GeV, $p_T^{\ell 4} > 1$ GeV are considered.

Different contributions are presented: exclusive, single-dissociative and double-dissociative. For modeling of single and double-dissociative contributions, two different photon PDFs are compared.

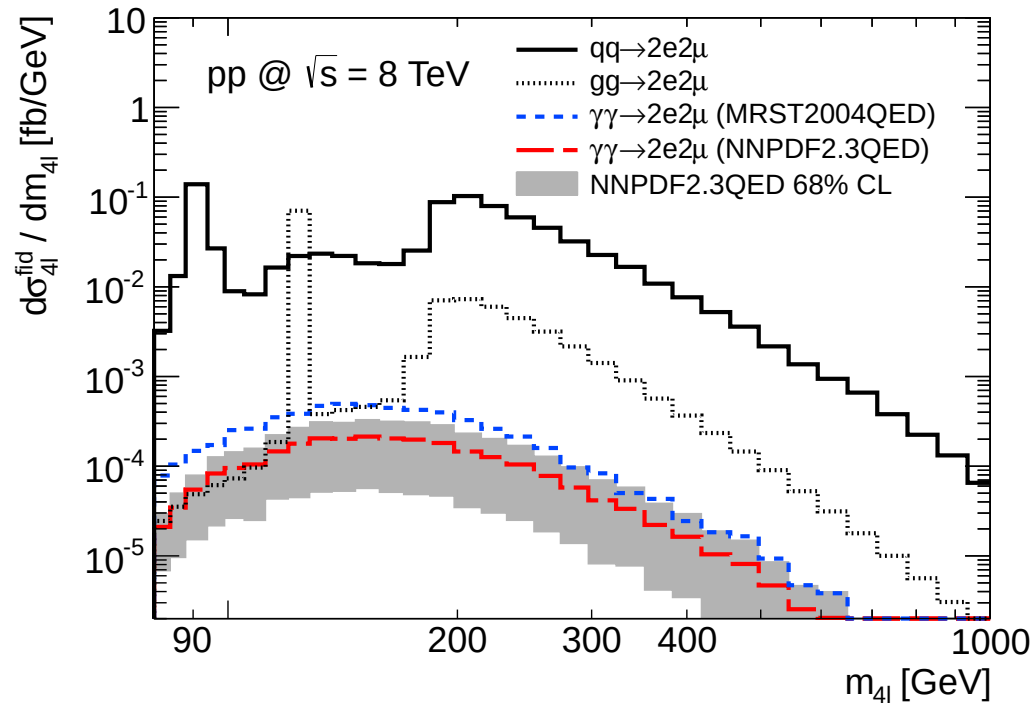
$\sqrt{s}$	$\sigma_{\gamma\gamma \rightarrow 4\ell}^{\text{excl}}$	$\sigma_{\gamma\gamma \rightarrow 4\ell}^{\text{sdiss}}$ with NNPDF2.3QED (MRST2004QED)	$\sigma_{\gamma\gamma \rightarrow 4\ell}^{\text{ddiss}}$ with NNPDF2.3QED (MRST2004QED)
8 TeV	0.22 fb	0.91 (1.4) fb	0.71 (2.0) fb
13 TeV	0.29 fb	1.2 (1.8) fb	0.86 (2.7) fb

Moreover, all quoted cross sections can be increased by a factor of 2.3 when the requirement on leading and subleading lepton p_T is lowered from 10 GeV to 5 GeV.

We confirm: [As a rule of thumb, the full cross section \(in any given kinematic range\) is about 10 times the pure elastic cross section.](#)

Comparing to SM processes at the LHC (including $H \rightarrow ZZ \rightarrow 4\ell$)

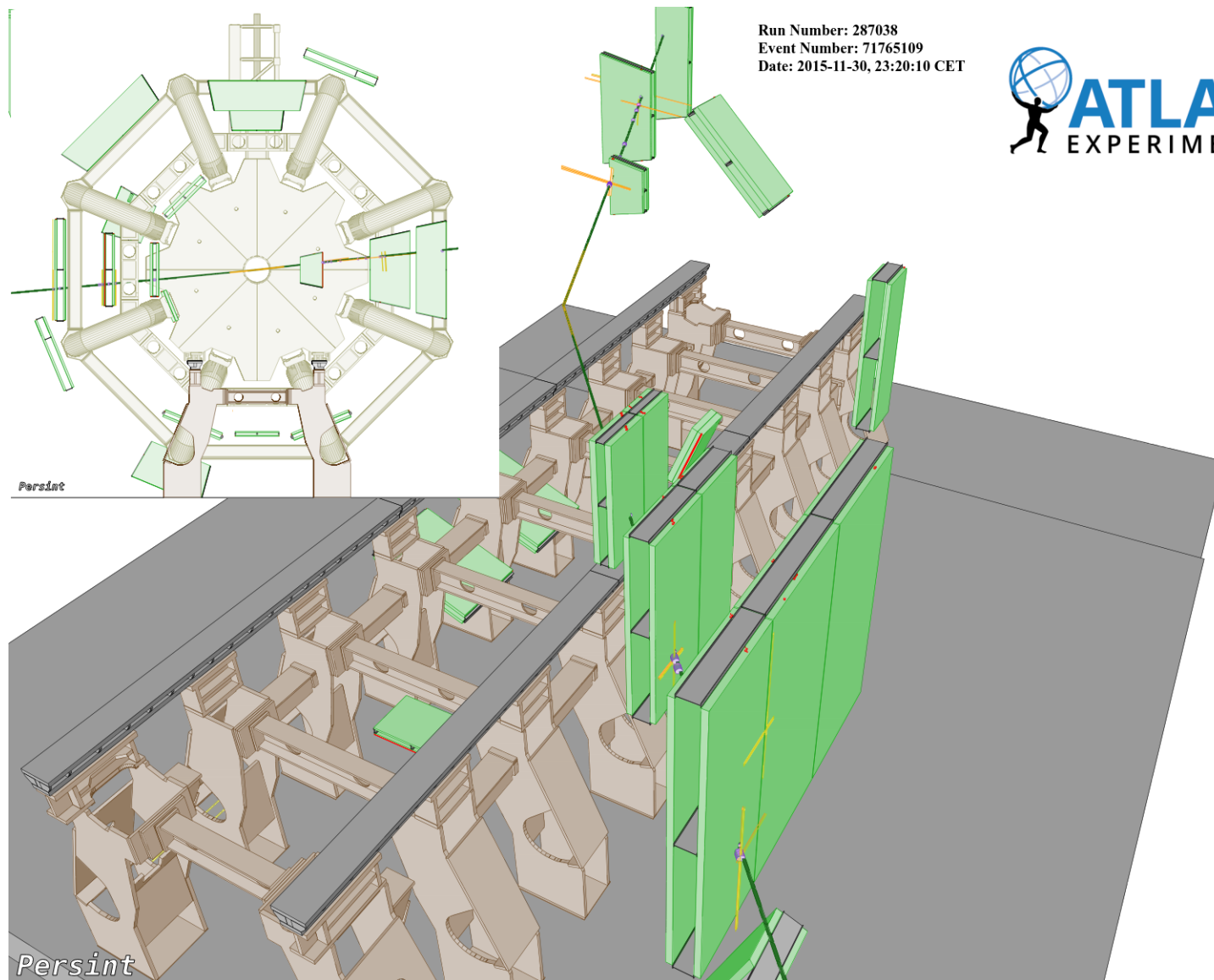
Differential cross sections ($d\sigma/dm_{4\ell}$) from the $\gamma\gamma$, $q\bar{q}$ and gg initial states at $\sqrt{s} = 8$ TeV for the $e^+e^-\mu^+\mu^-$ final state in the fiducial phase space from [arXiv:1509.07844]. For $\gamma\gamma$ production, two different photon PDFs are compared, where the shaded region indicates 68% confidence level for the NNPDF2.3QED PDFs.

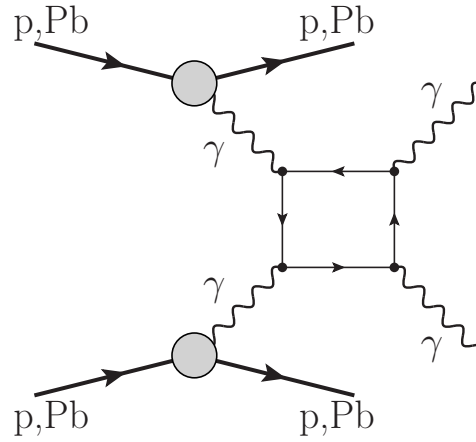


Chapter 5

$$Pb + Pb \rightarrow PbPb(\gamma\gamma) \rightarrow Pb + Pb + \gamma\gamma$$
$$(\gamma\gamma \rightarrow \gamma\gamma)$$

$$Pb + Pb \rightarrow PbPb(\gamma\gamma) \rightarrow Pb + Pb + \mu^+ + \mu^-$$





$$f_{\gamma/Pb}(\omega) = Z^2 \frac{\alpha_{em}}{\pi^2 \omega} \int d^2 \vec{k}_T \frac{k_T^2}{(k_T^2 + \frac{\omega^2}{\gamma^2})^2} G_E^2(k_T^2 + \frac{\omega^2}{\gamma^2}) \simeq Z^2 \frac{2\alpha_{em}}{\pi \omega} \int_0^{\frac{1}{R_{Pb}}} k_T dk_T \frac{k_T^2}{(k_T^2 + \frac{\omega^2}{\gamma^2})^2}$$

Then, $f(.)f(.)$ will be proportional to $Z^4 = 82^4 \simeq 5 \cdot 10^7$

With the enhancement factor (in Z^4) for $PbPb$, this cross section can become non negligible at the LHC (with standard running conditions).

We compute:

$$\sigma(PbPb(\gamma\gamma) \rightarrow Pb + Pb + \gamma\gamma) = S_{\gamma\gamma}^2 \cdot \int f_{\gamma/Pb}(\omega_1) f_{\gamma/Pb}(\omega_2) \sigma_{\gamma\gamma \rightarrow \gamma\gamma}(\omega_1, \omega_2) \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}$$

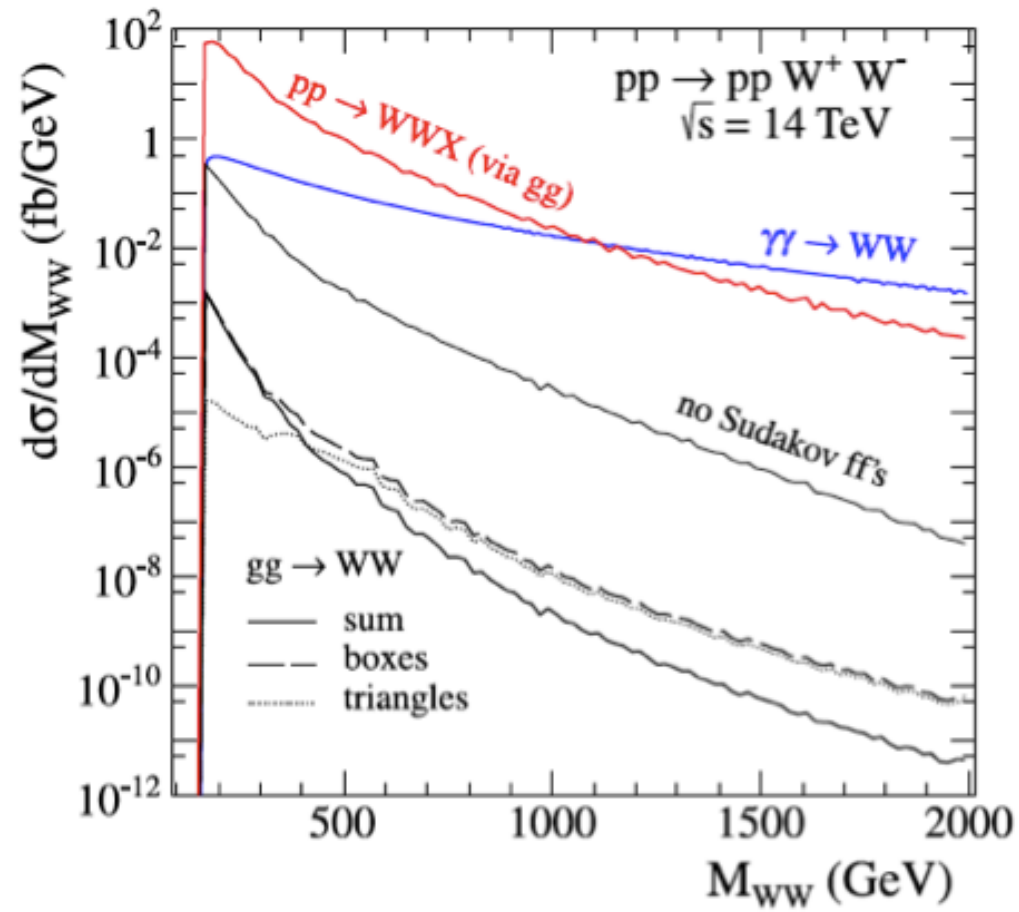
Important note: this is not possible to use Madgraph5 as a loop is needed in the $\gamma\gamma \rightarrow \gamma\gamma$.

With $\sqrt{s_{PbPb}} = 5.5$ TeV (standard), a first approximative calculation gives: $\sigma > 50$ nb for $W_{\gamma\gamma} > 5$ GeV and pseudo-rapidities < 2.5 , sharp dependence with these cuts (*calculations in progress*).

The main backgrounds (which may dominate the signal): (1) $gg \rightarrow \gamma\gamma$ and (2) $\gamma\gamma \rightarrow 2\ell$ (which then convert in photons) could be controlled with a proper experimental analysis.

In general

- The interest of studying a process mediated by a $\gamma\gamma$ sub-process is the clean signature of the events (once and only once the backgrounds are understood).
- Of course, tagging protons (for pp collisions) would be a plus.
- Example: **photon-induced neutralino pair production**. This would be an alternative to standard searches for such processes, with similar cross sections expected at large $W_{\gamma\gamma} \geq 0.5$ TeV [arXiv:1411.2920]...
- *In general, when $W_{\gamma\gamma}$ is large enough, $\gamma\gamma$ cross sections become competitive with standard processes... $\rightarrow$*



A final remark

- All along the discussion, we have used the so-called number of equivalent photons, directly linked to the intensity of the EM field... Obviously, according to the second quantization, all the discussion holds in classical physics or QM (almost equivalently). As the full discussion was done in special relativity (SR), it also holds in $\text{QM} + \text{SR} = \text{QFT}$ (quantum field theory).
- Therefore, what we have presented is very general... and we have checked this by the proof of equivalence with pure QED.
- However, we have never touched the wave function of the photons entering the interaction. A priori, this could have been considered as an option to solve the original problem.
- Why not doing so?

-
- The core of the problem is the definition of the photon wave function.
 - This is due to the non-localizable nature of the photon and the consequent difficulty in defining the position operator for photons.
 - This means photon wave function cannot exist in position eigenstate(s).
 - However photon wave function in momentum representation is well established and its momentum operator is well defined (*next slide in symbols*).
 - This is a very long standing problem in physics!
 - In our case, the photon is only quasi-real and also we have worked with a special case of EM field, localized in one coordinate of space, which means that we can define a mixed representation (space and momentum) for which a wave function would be defined (and the story continues)...
 - Therefore, this is a simpler problem in this big category. In practice, we could have worked directly with (quasi-real) photon wave functions... with the same results.

Mathematical elements on the photon wave function(s)

For simplicity, we consider one photon. We start with a standard QED expression:

$$|1ph\rangle = \sum \int d^3\vec{k} \phi(\vec{k}, s) a^\dagger(\vec{k}, s) |0\rangle .$$

This means that one photon is created, at given momentum $\vec{k}$. In order to be able to construct (as much as possible) a localized state in position space, we consider a wave-packet (superposition of several momenta). Each one has its own amplitude $\phi(\vec{k}, s)$ for a given polarisation s .

All this is well defined in QED and there is no mystery at this level.

The difficulty comes when we try to understand the idea of locality for one photon.

Position of the problem:

Let us define the vector function $\vec{\Phi}(\vec{r}, t)$ as: $\vec{\Phi}(\vec{r}, t) = \sum \int d^3\vec{k} \phi(\vec{k}, s) \vec{\epsilon}_{\vec{k}, s} e^{i(\vec{k}\vec{r} - \omega t)}$ where $\epsilon_{\vec{k}, s}$ is the (complex) polarization vector.

Then, the modulus squared of $\vec{\Phi}(\vec{r}, t)$, integrated over a volume dV gives the probability of locating the photon within the volume dV .

However, the vector function $\vec{\Phi}(\vec{r}, t)$ does not transform locally under Lorentz transformations. Moreover, there is no procedure to measure it. What can be measured are quantities like energy or the photoelectric detection probability of the photon.

Our purpose is to give some ideas why neither of them are localized in the volume into which $\vec{\Phi}(\vec{r}, t)$ has been confined.

In fact, this is very simple... The average photon energy is given by (with $\hbar = 1$):

$$\langle 1ph|H|1ph \rangle = \sum \int d^3\vec{k} \, \omega |\phi(\vec{k}, s)|^2$$

Then, let us define a new vector function: $\vec{\Phi}_E(\vec{r}, t) = \sum \int d^3\vec{k} \, \sqrt{\omega} \, \phi(\vec{k}, s) \, \vec{\epsilon}_{\vec{k},s} e^{i(\vec{k}\vec{r}-\omega t)}$.

Obviously: $\langle 1ph|H|1ph \rangle = \int d^3\vec{r} \, |\vec{\Phi}_E(\vec{r}, t)|^2$.

$\vec{\Phi}_E(\vec{r}, t)$ is like an energy wave function and $|\vec{\Phi}_E(\vec{r}, t)|^2$ plays the role of an energy density. But, this energy density is not locally connected with the photon density: $|\vec{\Phi}(\vec{r}, t)|^2$, and this is the main point to understand.

Indeed, from the Fourier expansions of $\vec{\Phi}_E(\vec{r}, t)$ and $\vec{\Phi}(\vec{r}, t)$ we find that they are related through (with the help of the convolution theorem):

$$\vec{\Phi}_E(\vec{r}, t) = \int d^3\vec{r}' \, G(\vec{r} - \vec{r}') \, \vec{\Phi}(\vec{r}', t)$$

where $G(\cdot)$ is a complicated function. Mathematically, this is because $G(\cdot)$ is not a $\delta(\cdot)$ fonction that the problem of non-locality of the photon wave function exists (and nothing else). All calculations done, we find:

$$G(\vec{r}) = \frac{1}{8\pi^3} \int d^3\vec{k} \, \sqrt{k} \, e^{i\vec{k}\vec{r}}.$$

Because of the spread associated with $G(\vec{r})$, $\vec{\Phi}_E(\vec{r}, t)$ can be non-zero at positions where $\vec{\Phi}(\vec{r}, t)$ is zero.

Appendix: probability of non-elastic interactions

The idea in this appendix is to rephrase the suppression factor:

$$S_{\gamma\gamma}^2 = \frac{\int_{b_{1,2} > R_p, |\vec{b}_1 - \vec{b}_2| > 2R_p} n(\vec{b}_1, \omega_1) n(\vec{b}_2, \omega_2) d^2\vec{b}_1 d^2\vec{b}_2}{\int_{b_{1,2} > 0} n(\vec{b}_1, \omega_1) n(\vec{b}_2, \omega_2) d^2\vec{b}_1 d^2\vec{b}_2}$$

in another form:

$$S_{\gamma\gamma}^2 = \frac{\int_{b_1 > r_p} \int_{b_2 > r_p} n(\vec{b}_1, \omega_1) n(\vec{b}_2, \omega_2) P_{non-inel}(|\vec{b}_1 - \vec{b}_2|) d^2\vec{b}_1 d^2\vec{b}_2}{\int_{b_1 > 0} \int_{b_2 > 0} n(\vec{b}_1, \omega_1) n(\vec{b}_2, \omega_2) d^2\vec{b}_1 d^2\vec{b}_2}.$$

Thus, replacing the non-overlapping condition $|\vec{b}_1 - \vec{b}_2| > 2r_p$ by a function $P_{non-inel}(|\vec{b}_1 - \vec{b}_2|)$ that would play the same role (to ensure the coherence of the reaction).

This means that $P_{non-inel}(|\vec{b}_1 - \vec{b}_2|)$ is a function that represents the probability that there is no interaction (no overlap) between the two colliding protons in impact parameter space.

This looks complicated. But, there is something we know from the elastic (measured) cross section: the proton-proton elastic scattering amplitude usually written as $\Gamma(b)$.

Then, very simply: $P_{non-inel}(b) = |1 - \Gamma(b)|^2$. Indeed, at a given center-of-mass energy, we know that if $\Gamma(b)$ is the elastic scattering amplitude, then the elastic cross section reads

$$\sigma_{el} = \int d^2\vec{b} |\Gamma(b)|^2$$

and the inelastic cross section reads

$$\sigma_{inel} = \int d^2\vec{b} [1 - |1 - \Gamma(b)|^2] .$$

Therefore, the combination $|1 - \Gamma(b)|^2$ can be interpreted as the probability for no inelastic interaction in a proton-proton collision at impact parameter b .

A parameterization exists for $\Gamma(b)$, which leads to:

$$P_{non-inel}(b) = |1 - \exp(-b^2/(2B))|^2,$$

where the value of $B = 19.7 \text{ GeV}^{-2}$ is taken from a measurement at $\sqrt{s} = 7 \text{ TeV}$ by the ATLAS experiment. At $\sqrt{s} = 13 \text{ TeV}$, we can use the extrapolated value $B = 21 \text{ GeV}^{-2}$.

Then, in practice, we can use $P_{non-inel}(b)$ or $\Theta(b - 2r_p)$ to quantify a non-overlapping condition between both protons.